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Sitting movement in elderly subjects with and without Parkinson's disease: A biomechanical study

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ABSTRACT

The aim of the present study was to compare kinetic, kinematic, and electromyographic variables during the sitting movement between healthy elderly and in those with Parkinson's disease (PD) with moderate involvement. We hypothesized that subjects with PD would show difficulty in selecting the muscles for the task and that this could be related to the co-activation pattern and would be reflected in the behavior of some biomechanical variables. Fifteen subjects participated in this study, seven healthy subjects (NN group) and eight with Parkinson's disease. Electromyography (EMG) activity of the tibialis anterior (TA), soleus (SO), vastus medialis oblique (VMO), biceps femoris (BF), and erector spinae (ES) were recorded, and biomechanical variables were calculated, during four phases of the sitting movement. Compared to healthy subjects, the subjects with PD showed more flexion at the ankle, knee, and hip joints in the initial position and lower joint velocity. However, the EMG activity and hip, knee, and ankle joint torques were not different during all phases of movement. The sitting movement in PD subjects with moderate involvement generates EMG activity and joint torques similar to healthy elderly subjects. Only a reduced movement velocity was found in PD patients during the sitting task.

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1. Introduction

The ability to sit down is a very important task for an individual's independence and can be used to evaluate functional level (Hase et al., 2004). Sitting down involves position changes for almost all corporal segments and requires satisfactory control over posture and balance so that the displacement of the center of mass is restricted. Although only a few studies have explored kinematic, kinetic and/or electromyographic aspects of this task (Ashford and De Souza, 2000; Kerr et al., 1997; Yoshida et al., 1983; Kralj et al., 1990; Mourey et al., 1998) they have, nevertheless, provided normative data on pattern of muscle activation, de-activation and joint movement.

Parkinson's disease (PD) promotes postural instability; this aspect of the disease has been widely explored (Stack et al., 2005; Frank et al., 2000; Van der Burg et al., 2006). Sitting, standing, reaching, and other common activities of daily life are associated with falls in people with PD (Wielinski et al., 2005; Bloem et al., 2004; Grimbergen et al., 2004; Cano-de la Cuerda et al., 2004). In the presence of severe PD, subjects needed support when reaching high and they also had a shorter functional reach (Stack et al., 2005).

Several studies were designed to evaluate a sit-to-stand motion, looking at center of mass displacement (Mourey et al., 1998; Inkster and Eng, 2004; Van der Burg et al., 2006), electromyography activities (Goulart and Valls-Solé, 1999; Bishop et al., 2005; Sande de Souza et al., 2011) and joint torques (Mak et al., 2003; Mak and Hui-Chan, 2005; Sande de Souza et al., 2011); few studies, however, have explored the sit task (Ashford and De Souza, 2000; Kerr et al., 1997; Yoshida et al., 1983; Kralj et al., 1990; Mourey et al., 1998). This is possibly due to the fact that the trajectories of all joints, in both tasks, for healthy subjects are similar, except that there is little visual information during sitting down (Mourey et al., 1998). The muscle activation patterns are also similar in these tasks, only differing in the type of contraction (Ashford and De Souza, 2000).

Observations in healthy subjects are not necessarily the same for subjects with diseases that affect postural stability, such as PD. Considering that PD subjects have a pattern of slowness of movement, muscle co-activation, and increased stiffness due to increased tonic activity of muscles (Dimitrova et al., 2004a), the biomechanical and EMG variables could have different patterns when compared to healthy subjects. Here, we hypothesized that subjects with PD can show difficulty in selecting the muscles for the task

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and that this could be related to a co-activation pattern and would be reflected in the behavior of some biomechanical variables. The present study can contribute to the literature adding information about the strategies used by PD subjects during the sitting task. The results could help in the decision making of a treatment proposal. The aim of the present study was to compare kinematic, kinetic and electromyographic variables during the sitting movement between healthy elderly and in those with PD, and to describe the movement strategies used by the PD subjects.

2. Methods

2.1. Subjects

Fifteen subjects participated voluntarily in the study. These included seven healthy subjects (NN group) and eight with Parkinson's disease (PD group) as shown in Table 1. For the subjects with Parkinson's disease, the mean duration of disease was 7.4 (±3.5) years. All subjects with PD showed bilateral involvement with the right side being the first affected, and were classified as stage 2 or stage 3 in the Hoehn and Yahr scale (Goetz et al., 2004). All PD subjects were medicated and undergoing physiotherapy during the period of the study. All subjects signed a consent form approved by the University Office of Research Ethics.

2.2. Experiment

All subjects were tested in the morning, after 12 h of medication withdrawal (*off* period). The anthropometric variables, specifically, height and weight, were measured after the tests.

The sitting tasks were done in a height-adaptable chair and the final position was established individually, in 60° of knee flexion (Fig. 1). This angle was selected because it requires a great quadriceps force level (Dionisio et al., 2008) and because the PD subjects were unable to perform a controlled sitting movement when the end position was more than 60° of knee flexion. The shoulders were kept by the subjects in 90° of flexion and the elbow was extended. The instructions were to look to a visual Target 1 in the upright position at the beginning and to perform the sitting until look to the Target 2, at the end of the movement, as is done for downward squatting (Cheron et al., 1997; Dionisio et al., 2008) and sit-to-stand (Sande de Souza et al., 2011). These targets were individually determined because of the different anthropometric variables and they were placed in the front of subjects on an acrylic board (Sande de Souza et al., 2011). The movement was performed with the right foot on the force plate and the left on a stable wooden platform; the subjects were instructed to maintain their feet in this position during the task, without any linear translation movement or axial rotation of the feet (Fig. 1). The subjects were also instructed to sit as quickly as possible, looking to the Target 1 in the beginning, and to the Target 2 at the end of movement, avoiding axial rotation. Ten trials were conducted after a training period of 3 trials.

Bipolar surface Delsys electrodes (Model DE2.2L) were placed on the muscle tibialis anterior (TA), soleus (SO), vastus medialis oblique (VMO), biceps femoris (BF), and erector spinae (ES) to register EMG activities. The electrodes were fixed after the skin surface was shaved, abraded, and cleaned with alcohol. For the VMO, the electrode was placed on the muscle belly, along the direction of the fibers (about 54°) and 2 cm distal from the motor point (Dionisio et al., 2008). To estimate the motor point localization, the subject was asked to perform an isometric contraction of the quadriceps so that the VMO muscle contour could be observed. For the other muscles, the electrodes were placed according to the guidelines of the SENIAM project (Surface ElectroMyoGraphy for the Non-Invasive Assessment of Muscles) from the Biomedical Health and Research Program (BIOMED II) of the European Union (Hermens et al., 2000). The EMG signals were amplified (2000×), band-pass filtered (20–450 Hz), and recorded at 1000 Hz.

The angular displacements of the hip, knee, and ankle joints, and the linear displacements of the shoulder, hip, knee, and ankle were calculated based on the *X* and *Y* coordinates, anterio-posterior and cephalo-caudal directions respectively, of the light emission diode (LED) markers placed at the hip (greater trochanter), knee (lateral epicondyle of the femur), ankle (lateral malleolus), and shoulder joint (lateral aspect of the acromion). In addition, markers were placed over the calcaneus, fifth metatarsal head, and the posterior corner of the force plate (Sande de Souza et al., 2011). The LED emissions were captured at a frequency of 100 Hz using a three-dimensional optical system (OPTOTRAK® 3020, Northern Digital Inc., Waterloo, Ontario, CA).

A force plate (AMTI OR6-5, Watertown, MA, USA) was used to record the ground reaction forces (*Fx*, *Fy*, and *Fz*) and the force moments (*Mx*, *My* and *Mz*) in orthogonal directions, at a sampling frequency of 1000 Hz. The signals were amplified (4000×), low-pass filtered (10.5 Hz), and recorded.

2.3. Data processing

Kinematic, kinetic, and EMG data were processed *"offline"* with a computer program using Matlab® software. The data processing calculated the linear displacements of the center of the ankle, knee, and hip joints, and the angular excursion of these joints. Also, these angles were differentiated to obtain angular velocity and acceleration of the joints. The antero-posterior position of the center of pressure (COP) was defined as the moment in the *y*-coordinate (*My*) divided by the vertical force (*Fz*). The COP locations in the anterior–posterior direction were reported as a percentage of the

Table 1
Descriptive data of the PD participants.

Subject	NN group				PD group				Disease duration (years)	Hoehn and Yahr	Medication
	Gender	Age (years)	Height (cm)	Weight (Kg)	Gender	Age (years)	Height (cm)	Weight (Kg)			
1	M	60	170	67	M	50	176	82	6	2	Prolopa/Selegiline
2	M	71	169	75	F	59	160	54	13	3	Prolopa/Cinetol
3	F	63	154	76	F	70	165	73	4	2	Prolopa
4	F	70	157	70	M	81	168	81	4	2	Prolopa/Selegiline
5	F	60	162	105	M	62	170	63	5	2	Prolopa
6	F	62	150	61	F	70	165	51	11	3	Prolopa
7	M	60	158	74	M	58	164	65	6	2	Prolopa
8	–	–	–	–	M	60	162	62	10	2	Prolopa
Mean (SD)	–	63,7(4.8)	160(7,4)	75.4(14.1)	–	63.8 (9.5)	166.3(5.0)	66.4 (11.5)	7.4 (3.5)	–	–

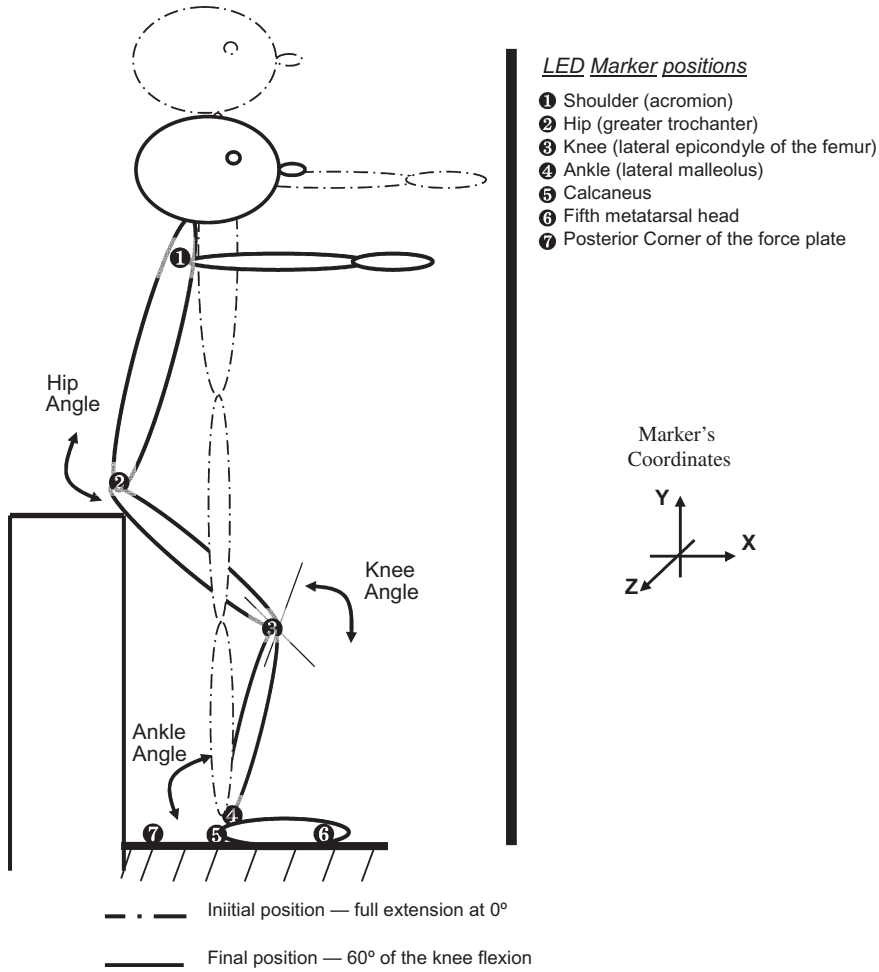

Fig. 1. This figure shows the initial and final positions of the sitting task. The LED marker positions, the ankle, knee and hip angles are also shown.

longitudinal foot length, from the most posterior tip of the heel to toe tip, of each subject.

The anthropometric data, including foot, leg, shank, and trunk lengths, were obtained from the *X* and *Y* coordinates of the marks placed at the center of each joint. The center of mass and moment of inertia of each segment were calculated based on the weight and sex of each subject using Zatsiorsky's model modified (de Leva, 1996). The joint torques of the hip, knee, and ankle were normalized to each subject's weight. The EMG signals collected during the movements were rectified, low pass filtered (at 25 Hz using a second-order Butterworth filter) and normalized to the average EMG signal recorded for the tested muscle during maximum voluntary isometric contraction (MVIC). The average EMG of the MVIC was calculated within the 500–1000 ms (ms) interval from the beginning of the isometric contraction.

The mean of the integrated EMG activities (iEMG), COP and joint torques (weight normalized) were calculated for each of the four phases which were based on the knee angular velocity: before the beginning of the movement (Phase 1, or P1), acceleration, (Phase 2, or P2) deceleration (Phase 3, or P3) and after the end of movement (Phase 4, or P4). Phase 1 was calculated for an interval of 100 ms, just before the knee velocity first achieved 5% of its peak, characterizing the pre-standup period. Phases 2 and 3 defined the acceleration and the deceleration time of the task, and included the interval from the end of Phase 1 to the time point where knee velocity achieved its peak and from the end of Phase 2 to the time point where knee velocity returned to 5% of its peak, respectively. Phase 4 defined the time for which the body remained at the target position, for a time interval of 100 ms, following the end of Phase 3 (Dionisio et al., 2008; Sande de Souza et al., 2011).

2.4. Statistical analysis

STATISTICA for Windows (StatSoft, Inc., Tulsa, OK; version 5.0) was used for analysis. Descriptive statistics (*t tests)* were used to test the effect of the group for linear and angular displacements. ANOVA with repeated measures design was used to test the effect of the groups (independent variables) and of the movement Phases (1–4) (independent variable) on the major dependent variables: the average values of the ankle, knee and hip joint torques, COP and the EMG signals from the recorded muscles (mean of each muscle separately). Also, the ANOVA was used to test the effect of groups and joints (hip, knee and ankle) on the maximum joint velocity. A post hoc comparison using the Tukey "honest significance test" was conducted to additional comparisons between groups. Alpha was set at 0.05.

3. Results

3.1. Temporal series

The temporal series reveals that the muscle pattern and joint torques during the phases of the movement (sitting movement) are similar among the NN and PD subjects (Figs. 2 and 3); the main differences were observed during the acceleration and deceleration

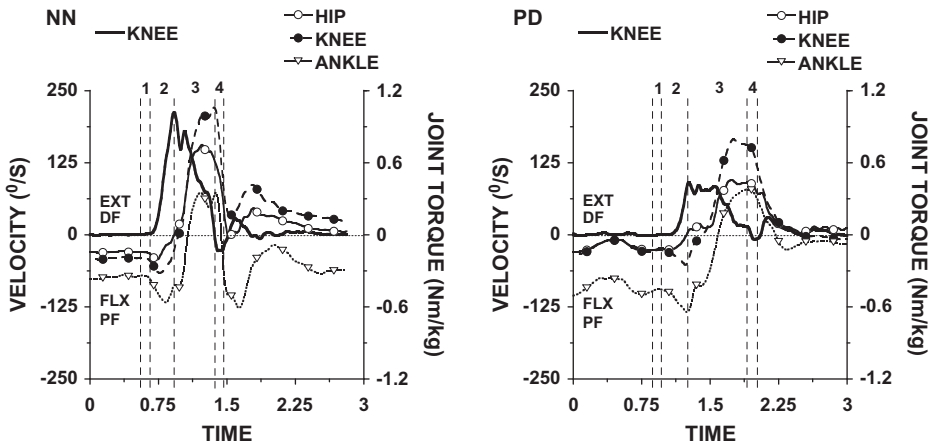

Fig. 2. This figure depicts the temporal series of the knee velocity (°/s) and joint torques (N m/kg) of the hip, knee, and ankle during sitting performed by one healthy subject (NN) and one with Parkinson's disease (PD). Vertical dotted lines represent the four phases of the movement. The positive values indicate extensor joint torque (EXT = extension) for hip and knee, and dorsiflexor joint torque for ankle (DF = dorsiflexion); while negative values indicate flexor joint torque (FLX = flexion) for hip and knee, and plantar flexion joint torque for ankle (PF = plantar flexion).

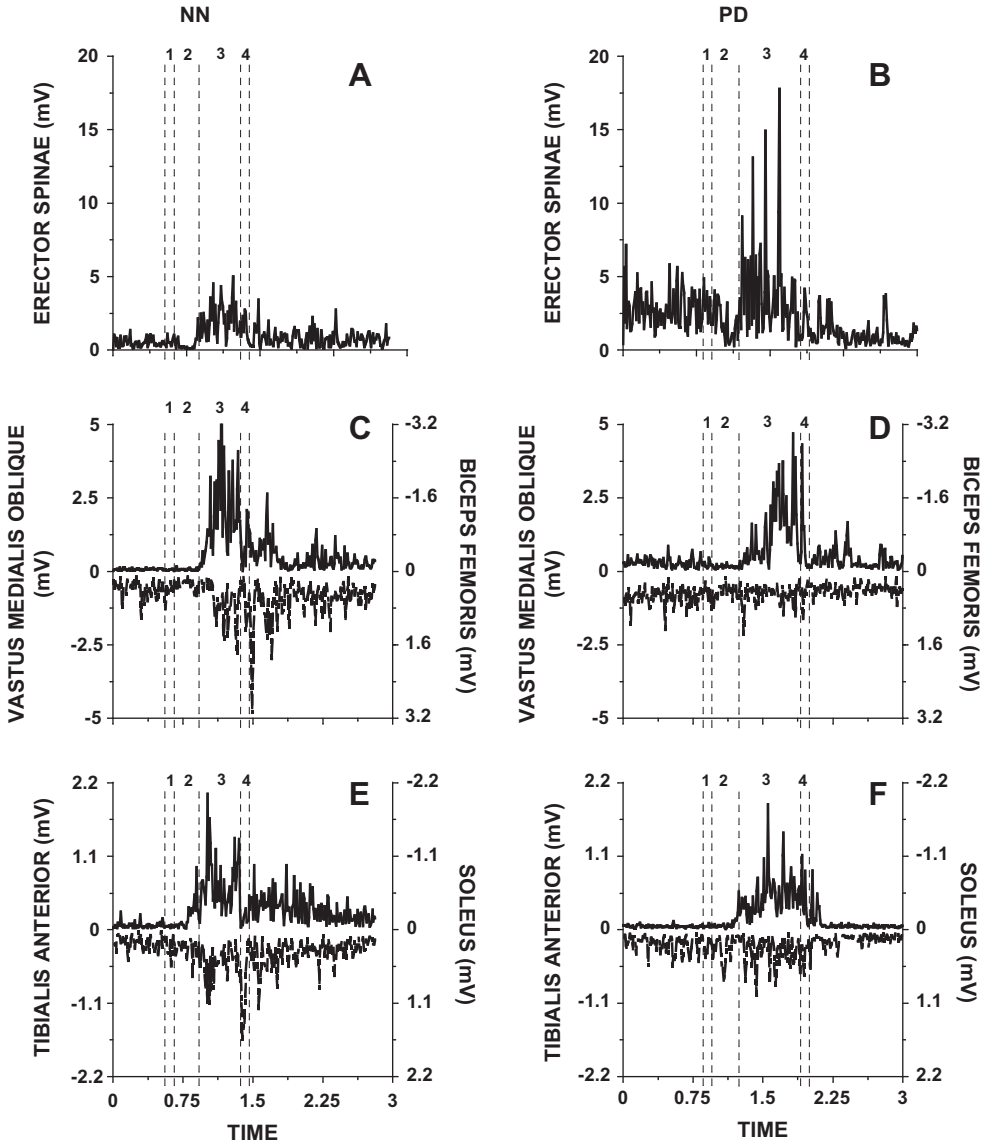

Fig. 3. This figure depicts the temporal series of muscle activities (mV) of the Erector Spinae (A and B), vastus medialis oblique and biceps femoris (C and D) and tibialis anterior and soleus (E and F) during sitting performed by one healthy subject and one with Parkinson's disease in the four phases of the movement (vertical dotted lines).

Table 2
The means and standard deviations (SD) of all subjects for maximal linear displacement measured in centimeters (cm), and angular displacement (degrees).

Variable	NN group (n = 7) Mean ± SD	PD group (n = 8) Mean ± SD	t Value	P value
Maximal liner displacement				
Shoulder AP	36.68 ± 13.42	27.49 ± 13.35	−1.32	0.2079
Shoulder CC	27.20 ± 3.41	30.33 ± 9.95	0.79	0.4436
Shoulder ML	3.13 ± 1.26	2.05 ± 0.90	−1.93	0.0755
Hip AP	35.24 ± 11.57	29.51 ± 8.17	−1.12	0.2829
Hip CC	25.32 ± 2.52	23.90 ± 5.95	−0.58	0.5684
Hip ML	3.41 ± 2.15	2.01 ± 1.17	−1.59	0.1352
Knee AP	9.17 ± 2.83	3.75 ± 1.54	−4.68	0.0004*
Knee CC	3.30 ± 1.40	2.30 ± 0.68	−1.79	0.0967
Knee ML	2.92 ± 2.13	2.93 ± 1.74	0.01	0.9934
Ankle AP	0.34 ± 0.09	0.23 ± 0.17	−1.48	0.1608
Ankle CC	0.44 ± 0.14	0.31 ± 0.17	−1.94	0.0735
Ankle ML	0.31 ± 0.07	0.37 ± 0.35	0.38	0.7063
Angular displacement				
Hip IP	6.67 ± 3.82	15.96 ± 9.14	2.49	0.0268*
Knee IP	7.12 ± 2.19	18.34 ± 8.22	3.48	0.0040*
Ankle IP	63.54 ± 7.00	71.26 ± 6.33	2.24	0.0430*
Hip FP	79.31 ± 4.86	81.95 ± 22.15	0.30	0.7631
Knee FP	65.61 ± 10.52	62.82 ± 4.45	−0.68	0.5048
Ankle FP	75.68 ± 9.49	75.37 ± 3.83	−0.08	0.9324
Hip AE	69.97 ± 7.84	65.99 ± 17.64	−0.55	0.5915
Knee AE	56.02 ± 12.76	43.23 ± 9.52	−2.21	0.0449*
Ankle AE	12.65 ± 4.59	5.35 ± 2.55	−3.87	0.0019*

Abbreviations: AP, antero-posterior; CC, cephalo-caudal; ML, medio-lateral; IP, initial position; FP, final position; AE, angular excursion.
* Statistical significance, $P < 0.05$.

phases. The PD subject presented longer duration of the acceleration and deceleration phases compared to the NN subject. In the acceleration phase, both subjects showed ankle joint torque in plantar flexion, while knee and hip joint torques were in flexion. In this phase, the EMG activity of the tibialis anterior was pronounced in the NN subject (Fig. 3E and F). The EMG activity of the erector spinae presented a partial and more slowed decrease in the acceleration phase in PD subject. In the deceleration phase, for knee joint torque the PD subject presented a lower value (Fig. 2). All muscles presented the maximum magnitude in the acceleration phase, and a co-activation pattern was revealed between the tibialis anterior and soleus, vastus medialis oblique and biceps femoris (Fig. 3C–F) for both subjects. Also in this phase, in a qualitative analysis, it can be observed that the erector spinae activity was greater in PD subject (Fig. 3A and B).

3.2. Linear and angular displacements

The results indicated that PD subjects did not perform the sitting task in the same way as healthy subjects (Table 2). The subjects with PD produced smaller linear displacement of the knee in the antero-posterior (AP) direction ($p = 0.0004$; Fig. 4C). Although there were no statistical differences in other directions ($p > 0.07$), the PD group presented a tendency to produce smaller linear displacement of shoulder and hip joints in the AP direction (Fig. 4A–D). The initial position or the initial joint angle adopted was different for all joints ($p < 0.05$). In the final position, both groups were similar ($p > 0.50$). Also, there were statistical differences in the angular excursion (AE) of knee and ankle joints ($p < 0.05$), but for hip joints the AE was not different between groups ($p = 0.59$) (Fig. 4E–G).

3.3. COP, Ankle, Knee and Hip Joint Torques

For COP ($F_{(1,13)} = 0.13$; $p < 0.72$), ankle ($F_{(1,13)} = 0.00$; $p < 0.99$), knee ($F_{(1,13)} = 2.88$; $p < 0.11$), and hip ($F_{(1,13)} = 0.90$; $p < 0.36$) joint torques (Fig. 5A–D), the ANOVA did not reveal any effects for the groups of subjects. Although all variables presented effects for the phases of the movement ($F_{(3,39)} = 6.00$; $p < 0.001$; $F_{(3,39)} = 22.76$; $p < 0.001$; $F_{(3,39)} = 29.79$; $p < 0.001$; $F_{(3,39)} = 40.60$; $p < 0.001$), respectively for COP, ankle, knee and hip joint torques. The interaction between these factors was not observed for COP ($F_{(3,39)} = 1.48$; $p < 0.23$), ankle ($F_{(3,39)} = 0.52$; $p < 0.66$), knee ($F_{(3,39)} = 0.30$; $p < 0.82$) and hip ($F_{(3,39)} = 0.05$; $p < 0.98$) joint torques.

3.4. Electromyography

For all muscles the EMG activity was different in each of the four movement phases for both groups (TA – $F_{(3,39)} = 14.43$; $p < 0.001$; SO – $F_{(3,39)} = 3.24$; $p < 0.03$; VMO – $F_{(3,39)} = 9.50$; $p < 0.001$; EE – $F_{(3,39)} = 7.87$; $p < 0.001$)(Fig. 6A and C–F), except for BF ($F_{(3,39)} = 1.27$; $p < 0.29$). However, there were no effects for the subject groups for TA ($F_{(1,13)} = 0.61$; $p < 0.44$), SO $F_{(1,13)} = 3.62$; $p < 0.07$), VMO ($F_{(1,13)} = 0.00$; $p < 0.99$), BF ($F_{(1,13)} = 1.54$; $p < 0.23$) and EE ($F_{(1,13)} = 0.07$; $p < 0.80$). The interactions were not significant $F_{(3,39)}<2.43$; $p > 0.07$).

3.5. Maximal joint velocity

The average of maximal velocity of the hip, knee, and ankle joints presented effects by group ($F_{(1,13)} = 4.69$; $p < 0.04$) and joints ($F_{(2,26)} = 103.57$; $p < 0.0001$), but did not presented interaction ($F_{(2,26)} = 2.52$; $p < 0.0998$). The post hoc analysis revealed group effects for hip ($p = 0.005$) and knee ($p = 0.0007$) joints, but not for the ankle ($p = 0.49$). Also, for both groups (Fig. 6B), the hip velocity was greater than the knee velocity, which in turn was greater than the ankle velocity ($p < 0.03$).

4. Discussion

The results revealed that the main difference between the groups was in the initial position, during which the PD group presented larger flexion of the joints compared to the NN group.

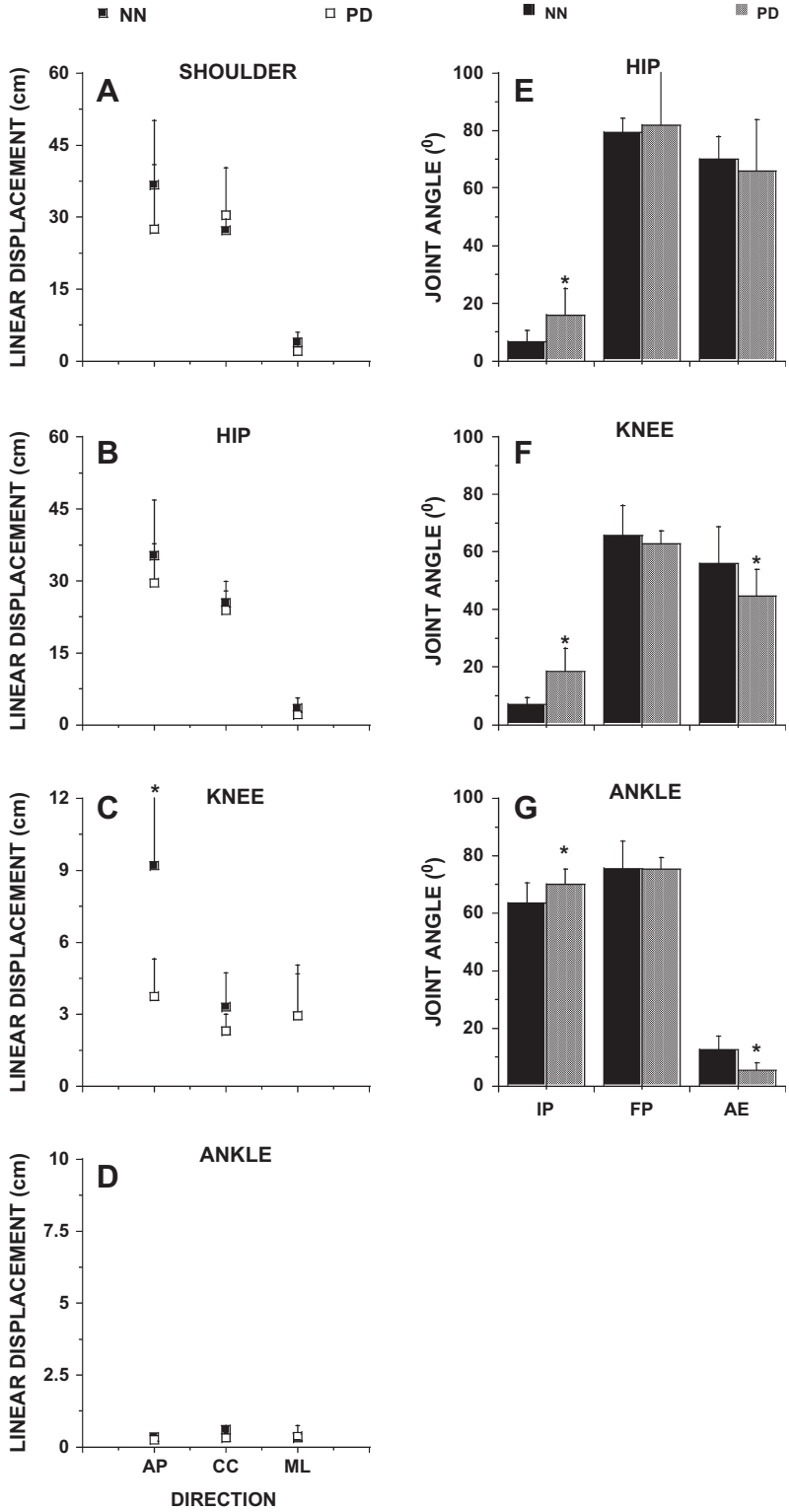

Fig. 4. This figure depicts the average and standard deviation across all subjects of the shoulder (A), hip (B), knee (C) and ankle (D) linear displacement; and hip (E), knee (F) and ankle (F) angular displacement during the sitting movement. (NN, closed squares; PD, open squares); AP: antero-posterior, CC: cephalo-caudal and ML: medio-lateral direction; IP: initial position; FP: final position; AE: angular excursion. Asterisk (*) indicates $p < 0.05$.

This condition affected the linear and angular displacements. As the PD group was in a more flexed position, the knee was already in an anterior position because of the dorsiflexion of the ankle and there was also lower linear displacement of the knee in the anterior direction. The PD group maintained this flexor pattern during movement execution, whereas the NN subjects were able to change angular positions according to the task necessities. We had hypothesized that subjects with PD would show difficulty in

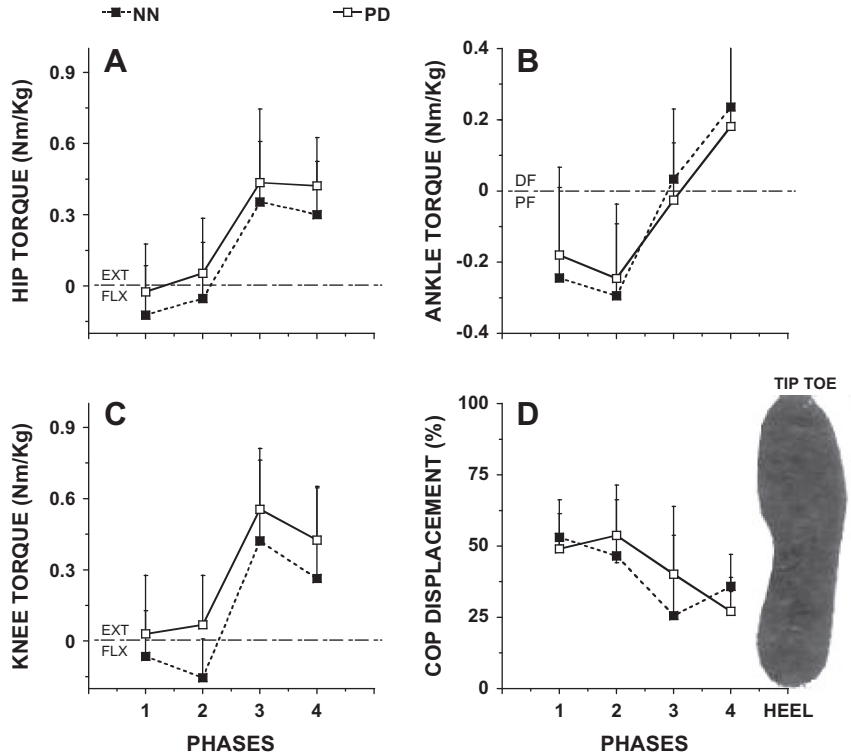

Fig. 5. This figure depicts the average and standard deviation across all subjects of the hip (A), ankle (B), and knee (C) joint torques (N m/kg) and COP position (D) during the four phases of sitting movement. Note that the COP position in the anterior–posterior direction was reported as a percentage of the longitudinal foot length (NN, closed squares; PD, open squares). EXT = extension; FLX = flexion; DF = dorsiflexion; PF = plantar flexion. Asterisk (*) indicates $p < 0.05$.

selecting the muscles for the task because of the co-activation tendency, as presented in many articles in different tasks. This co-activation would be reflected in the biomechanical variables. Our data did not support completely these hypotheses, since the joint torques and COP position did not present differences between groups for the sitting task (Fig. 5A–D). Also, in general, the EMG activities showed that the activities of the two subjects were similar between NN and PD subjects, during all phases of movements, although the statistics only suggested a non-significant tendency for SO. In qualitative terms, the EMG activities of BF, TA, and SO muscles were always larger for PD group (Fig 6C and D–F). This strategy can help the stability of the joints because the PD subjects maintained a flexed knee position during all phases of the movement. However, the EMG activities of the SO and TA did not produce group differences in the ankle joint torque. Among the muscles registered, the SO was responsible for plantar flexor ankle joint torque in the P1 and P2, but the TA was responsible for dorsiflexor ankle joint torque during P3 and P4. During all phases, these muscles were co-activated in both groups. Similar observations have been noted for downward squatting (Dionisio et al., 2008). Although, the co-activation did not differentiate the groups, we can speculate that an increase in the amount of EMG over time and/or the bradykinesia could contribute to the slower velocity observed in the PD group.

The BF helps pelvis stability during flexion of the hip and trunk (Sande de Souza et al., 2011) and its activity can be related to the greater velocity in the hip compared to the knee. This velocity on the hip and knee is influenced by the gravitational action, and the BF could control the hip flexion by an eccentric contraction, since hip velocity was larger than knee. The continuous EMG activity of the BF can be observed in both groups (Fig. 3C and D–C) and suggests that this muscle is important for the task. Also, BF activation helps to increase the knee joint torque, especially in the acceleration phase which begins with the flexion of the knee. Fig. 5C shows that knee joint torque for the PD group presented a tendency to the extensor pattern for all phases of the sitting movement. This tendency was also observed for hip joint torque, which was extensor in all phases of the movement (Fig. 5A). We believe that the BF activity was partially responsible for this pattern, although no statistical difference was observed. Although we did not test this, the bradykinesia seems to have affected the hip joint less than the knee joint (Fig. 6B) and the BF was important to control the movement of the pelvis. The greater hip mobility could lead to the tendency for a more forward COP position for the PD group (P2 and P3), as shown in Fig. 5D. This could cause greater EMG activity of the soleus and biceps femoris in the same phases, to maintain the COP in a safe base of support.

Elderly subjects have a lesser trunk velocity and more difficulty during sitting because the movement requires fine postural control and accuracy in the final part of the movement as compared to young subjects (Mourey et al., 1998). During sitting, elderly subjects needed greater velocity control to maintain their equilibrium in an upright initial position, as well as to achieve accuracy in making contact with the seat. Also the mechanical demand in the hip joint is greater than in knee and ankle joints (Flanagan et al., 2003). As well, several studies have suggested that healthy subjects have the ability to take initial biomechanical conditions into account to flexibly modify postural responses to maintain stability during postural changes (Macpherson et al., 1989). On the other hand, PD subjects have difficulty modifying the magnitude and patterns of postural responses for changes in postural demand. The importance of the basal ganglia for set-dependent adaptation of postural movement patterns for changes in conditions has been shown previously for changes in arm support, changes from a

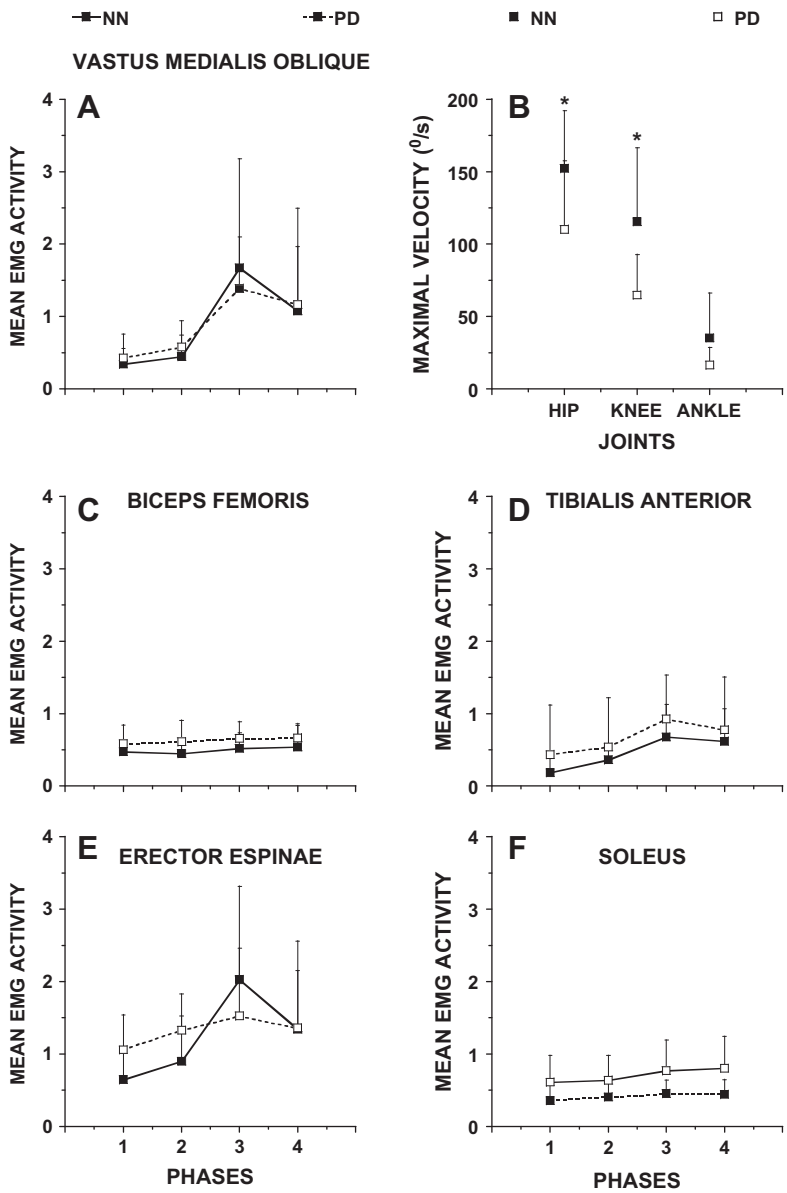

Fig. 6. This figure depicts the average and standard deviation across all subjects of the iEMG activities during each the four phases of the sitting movement for vastus medialis oblique (A), biceps femoris (C), tibialis anterior (D), erector spinae (E) and soleus (F). (B) depicts the maximal joint velocity for hip, knee and ankle for both groups (NN, closed squares; PD, open squares). Asterisk (*) indicates $p < 0.05$.

standing to sitting posture, and surface rotation versus translation (Chong et al., 1999; Horak et al., 1992). Difficulty in modifying the postural response can be related to axial rigidity. Several studies have shown increased axial rigidity in PD subjects, especially in the *off* state (Steiger et al., 1996; Van Emmerik et al., 1999). Although we did not measure rigidity, and explored only the ES muscle, it is possible that the rigidity in the *off* state could have contributed to the tendency to larger EMG activities (in qualitative terms) that we observed in BF, SO, and TA during the sitting task. Dimitrova et al. (2004a,b) observed that BF, SO, and TA associated with other muscles (tensor fascia latae, adductor longus and rectus abdominis) seem to control the body, helping balance recovery. Exploring different stance widths, they found that PD subjects do not modify the magnitude of their muscle activity to adjust to the stance widths. They concluded that PD subjects may lack trunk flexibility because of axial rigidity, reflecting in their co-contraction of hip and trunk muscles (Dimitrova et al., 2004a). For Jacobs et al. (2005) the biomechanical effects of a flexed posture of PD subjects further limit the amount of torque they can generate at the support surface, exaggerating bradykinetic postural responses. We did not find differences in torque generation; however, the bradykinesia presented in other studies, which explored automatic postural responses, was also observed in the present study (Figs. 3 and 6B), noting, however, that the sitting movement can be classified as automatic plus voluntary, involving trunk, upper, and lower limbs.

In a previous study, Sande de Souza et al. (2011) evaluated the biomechanical variables during a sit-to-stand movement with the trunk partially restricted. This study also observed similarities between healthy and PD groups for joint torque, COP displacement, and EMG activities. In the present study we used the same position of the arms and, therefore, the trunk was also partially restricted; in part, this could explain our results. As known, the basal ganglia can monitor initial conditions to modify existing motor programs

(Marsden, 1984). To flexibly modify postural responses for changes in biomechanical demands throughout the sitting task, a healthy nervous system needs accurate information from proprioceptive receptors about perturbation characteristics, as well as an accurate internal model of body biomechanics to accurately shape postural responses for consistent postural stability. The difficulty that PD subjects show in modifying their postural responses for biomechanical demands may be due to poor use of proprioceptive information (Zia et al., 2000) or to poorly developed internal representations of their bodies, the latter being necessary to customize motor programs (Horak and Macpherson, 1996). Thus, these results suggest that the basal ganglia are critical for adapting postural response patterns for specific biomechanical conditions (Chong et al., 1999; Horak and Macpherson, 1996).

Some authors have also recognized that voluntary movements are known to be weak in PD subjects (Kakinuma et al., 1998; Stelmach et al., 1989). The strength of the spine, hip, knee, and ankle muscles is important to maintain or to recover postural stability (Nocera et al., 2010). In particular, eccentric resistance training has been suggested to increase muscle hypertrophy (Allen et al., 2010), decrease the bradykinesia, and improve quality of life (Dibble et al., 2009). However, it can be noted that the PD subjects investigated in the present study (stage 2 and 3, Hoehn and Yahr) could perform the activity in a very similar way of healthy subjects, with exception for the lower velocity.

This study should be interpreted in the light of the f ollowing limitations: (1) sample size was relatively small and consisted of individuals with mild-to-moderate PD; (2) the task produced a partial restriction of the trunk and may have affected more the variables studied in the healthy group than the PD group. Moreover, this study reinforces the observations of other authors (Dimitrova et al., 2004a,b) about the importance of trunk muscle activation and generation of torques on the lower limbs.

5. Conclusion

The results of our study show that PD patients with moderate involvement were similar to elderly subjects in the performance of sitting movements. Only a reduced movement velocity was found in PD patients during the sitting task.

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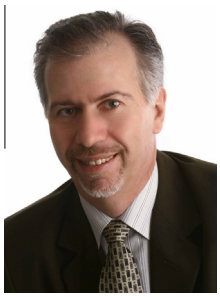

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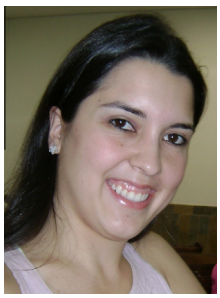

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